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Microsoft Windows [Version 10.0.26200.8894]
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C:\Users\vedad>mysql -u root -v
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: NO)

C:\Users\vedad>mysql -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> -- STEP 1: CREATE DATABASE
mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql> USE bookflow_db;
Database changed
mysql>
mysql> -- STEP 2: CREATE BOOKS TABLE
mysql> CREATE TABLE Books (
->     book_id INT PRIMARY KEY,
->     title VARCHAR(100) NOT NULL,
->     isbn VARCHAR(20) UNIQUE,
->     published_year INT CHECK (published_year < 2027)
-> );
ERROR 1050 (42S01): Table 'books' already exists
mysql>
mysql> -- STEP 3 & 4: INSERT BOOKS
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
-> (1, 'The Great Gatsby', '9780743273565', 1925),
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
```

```
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-> (3, '1984', '9780451524935', 1949);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
```

```
mysql>
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title       | isbn       | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988 |
| 2       | Clean Code   | 9780132350884 | 2008 |
| 3       | Atomic Habits | 9780735211292 | 2018 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- STEP 5, 6 & 7: CREATE MEMBERS TABLE
mysql> CREATE TABLE Members (
->     member_id INT PRIMARY KEY,
->     full_name VARCHAR(100),
->     email VARCHAR(100) UNIQUE
-> );
ERROR 1050 (42S01): Table 'members' already exists
mysql>
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name   | email               |
+-----+-----+-----+
| 1         | Anil Kumar  | anil.kumar@example.com |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma  | ravi.verma@example.com |
| 101       | John Smith  | john.smith@email.com |
| 102       | Emma Wilson | emma.wilson@email.com |
| 103       | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
mysql> -- STEP 8: CREATE LOANS TABLE
mysql> CREATE TABLE Loans (
  ->   loan_id INT PRIMARY KEY,
  ->   member_id INT,
  ->   book_id INT,
  ->   loan_date DATE,
  ->   FOREIGN KEY (member_id) REFERENCES Members(member_id),
  ->   FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- STEP 9 & 10: INSERT LOANS
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
  -> (1, 101, 1, '2025-01-05'),
  -> (2, 102, 2, '2025-01-08'),
  -> (3, 103, 3, '2025-01-10'),
  -> (4, 101, 2, '2025-02-01'),
  -> (5, 102, 1, '2025-02-05'),
  -> (6, 103, 2, '2025-02-12'),
  -> (7, 101, 3, '2025-03-01'),
  -> (8, 102, 3, '2025-03-07'),
  -> (9, 103, 1, '2025-03-15'),
  -> (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.00 sec)
```

```
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> -- STEP 9 & 10: INSERT LOANS
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'),
-> (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'),
-> (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'),
-> (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'),
-> (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'),
-> (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
mysql> -- STEP 11: JOIN QUERY
```

```
mysql>
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql>
mysql> -- STEP 11: JOIN QUERY
mysql> SELECT m.full_name AS Member_Name,
->      b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m
-> ON l.member_id = m.member_id
-> INNER JOIN Books b
-> ON l.book_id = b.book_id;
```

```
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| John Smith | The Alchemist |
| Emma Wilson | The Alchemist |
| Michael Brown | The Alchemist |
| John Smith | The Alchemist |
| Emma Wilson | Clean Code |
| John Smith | Clean Code |
| Michael Brown | Clean Code |
| Michael Brown | Atomic Habits |
| John Smith | Atomic Habits |
| Emma Wilson | Atomic Habits |
+-----+-----+
```

```
+-----+
10 rows in set (0.01 sec)

mysql>
mysql> -- STEP 12: GROUP BY QUERY
mysql> SELECT published_year,
->      COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
+-----+
| published_year | Total_Books |
+-----+
|          1988 |          1 |
|          2008 |          1 |
|          2018 |          1 |
+-----+
3 rows in set (0.00 sec)

mysql>
mysql> -- STEP 13: CREATE DONATION HISTORY TABLE
mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> -- STEP 14-16: TRANSACTION
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql>
```

```
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
    -> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Donation_History
    -> (donation_id, book_id, donor_name, donation_date)
    -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> -- DISPLAY DONATION HISTORY
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
|          1 |        4 | Raj Kumar  | 2026-08-22    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> -- STEP 17: CREATE INDEX
mysql> CREATE INDEX idx_books_isbn
    -> ON Books(isbn);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> -- STEP 18: FAST SEARCH USING ISBN
mysql> SELECT *
    -> FROM Books
    -> WHERE isbn = '9780451524935';
Empty set (0.00 sec)
```

```
mysql> SELECT *  
      -> FROM Books  
      -> WHERE isbn = '9780451524935';  
Empty set (0.00 sec)
```

```
mysql>  
mysql> -- FINAL TABLES  
mysql> SELECT * FROM Books;  
+-----+-----+-----+-----+  
| book_id | title       | isbn       | published_year |  
+-----+-----+-----+-----+  
| 1 | The Alchemist | 9780061122415 | 1988 |  
| 2 | Clean Code   | 9780132350884 | 2008 |  
| 3 | Atomic Habits | 9780735211292 | 2018 |  
| 4 | Animal Farm  | 9780451526342 | 1945 |  
+-----+-----+-----+-----+  
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Members;  
+-----+-----+-----+  
| member_id | full_name   | email                |  
+-----+-----+-----+  
| 1 | Anil Kumar   | anil.kumar@example.com |  
| 2 | Priya Sharma | priya.sharma@example.com |  
| 3 | Ravi Verma   | ravi.verma@example.com  |  
| 101 | John Smith  | john.smith@email.com    |  
| 102 | Emma Wilson  | emma.wilson@email.com   |  
| 103 | Michael Brown | michael.brown@email.com |  
+-----+-----+-----+  
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Loans;  
+-----+-----+-----+-----+  
| loan_id | member_id | book_id | loan_date |  
+-----+-----+-----+-----+  
| 1 | 101 | 1 | 2025-01-05 |  
| 2 | 102 | 2 | 2025-01-08 |  
| 3 | 103 | 3 | 2025-01-10 |  
| 4 | 101 | 2 | 2025-02-01 |  
| 5 | 102 | 1 | 2025-02-05 |  
| 6 | 103 | 2 | 2025-02-12 |
```

```
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1 | 4 | Raj Kumar | 2026-08-22 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> |
```